

CS & IT ENGINEERING

OPERATING SYSTEM

Intro and Background

Lecture No. 3



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A blue laptop with an orange keyboard is shown from a slightly elevated angle. The screen displays the text 'TOPICS TO BE COVERED' in a dark blue, sans-serif font, centered within a white rectangular area that has a thin blue border.

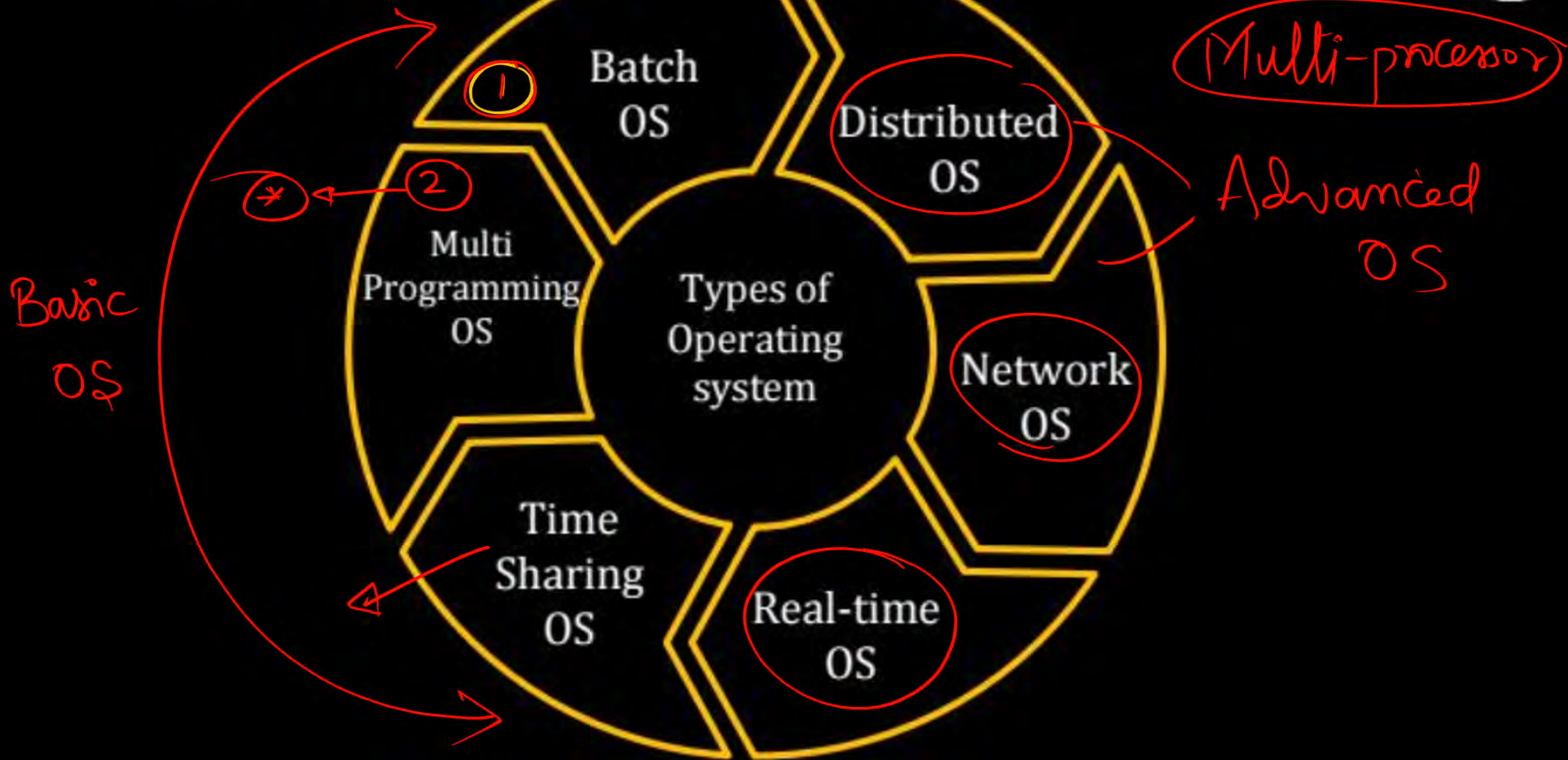
TOPICS TO
BE
COVERED

Types of OS

Arch. Tequirements

Mode Shifting *

Types of O.S



1st Gen of Computers (1930-40's) : No OS

2nd Gen " " (1940-50's) : Magnetic Tapes :

→ No OS

3rd Gen of Computers (1950's-60's)

↓
Magnetic Disk (Hard disk ; Floppy disk)

(P.C)



UNI-PROGRAMMED

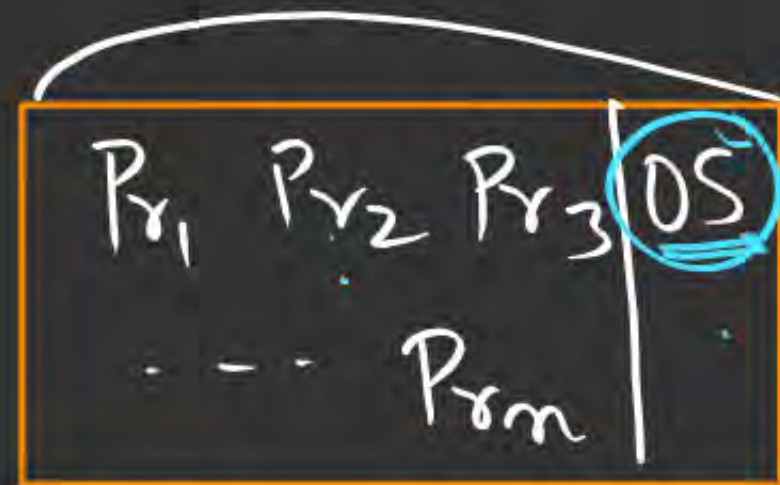


MULTI PROGRAMMED

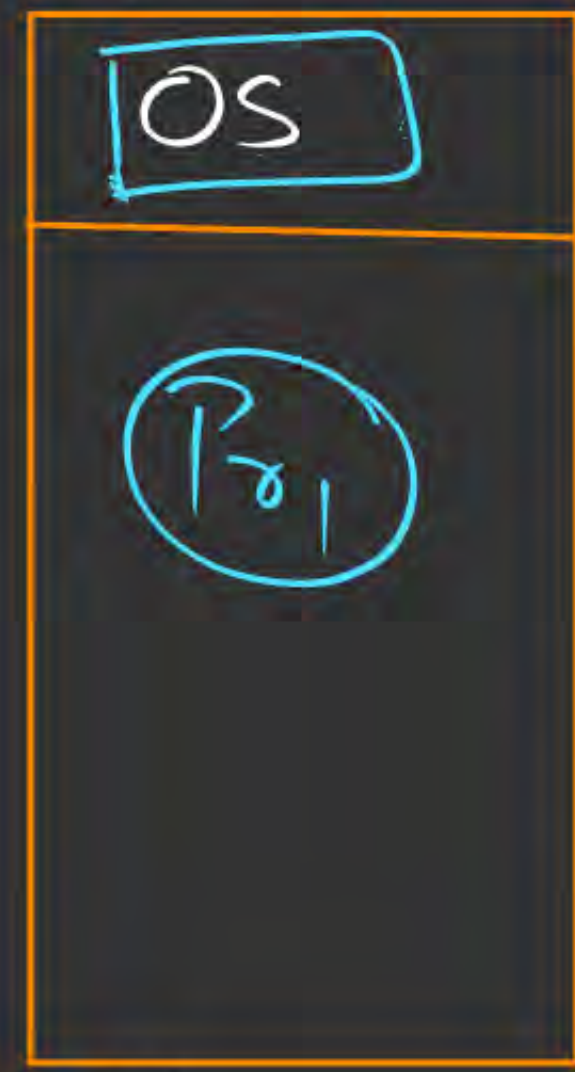
Uniprogramming: Ability of the O.S to hold a single program in memory,

IO: Input-output

IOD: IO-device



Hard-disk
(HDD)



Memory
(RAM)

S-Area
(System)

U-Area (user)
(Programs)

PC = Personal Computer

Stored Program Concept

Drawback of Uniprogramming: Ex: ^(Non-GUI) MS-DOS (disk O.S)

Single Program in Memory is not enough (sufficient) to keep the CPU always busy;

Idleness of CPU:

$$\text{Throughput} = \frac{\text{No. of Programs Completed}}{\text{Unit Time}}$$

→ Throughput will be less
(Efficiency)

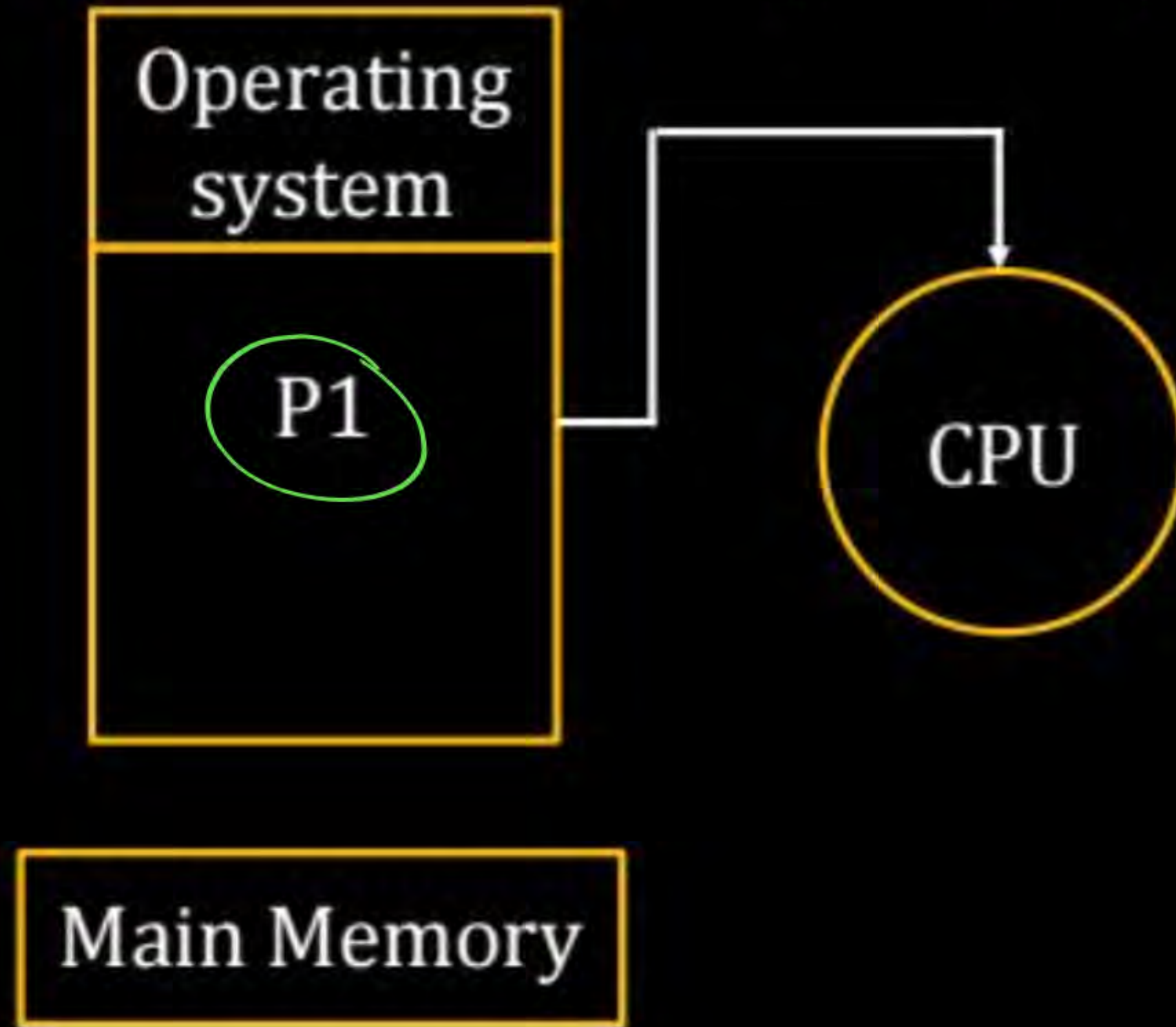
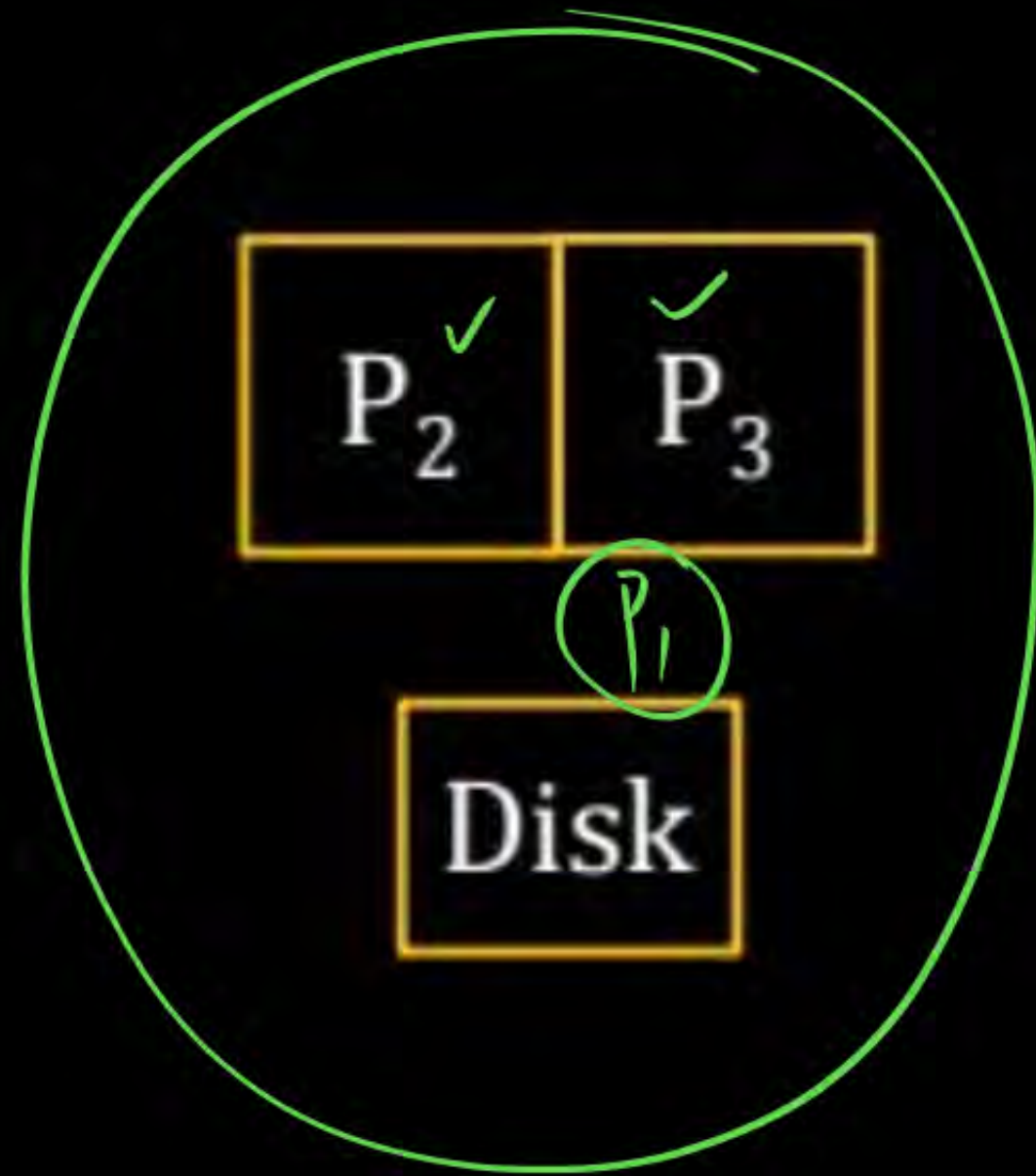
10 progs - 60's
? - 15

$$\frac{10}{60} = \left(\frac{1}{6}\right)$$

120 m - 20 topics
1 m - ?

$$\frac{20}{120}$$

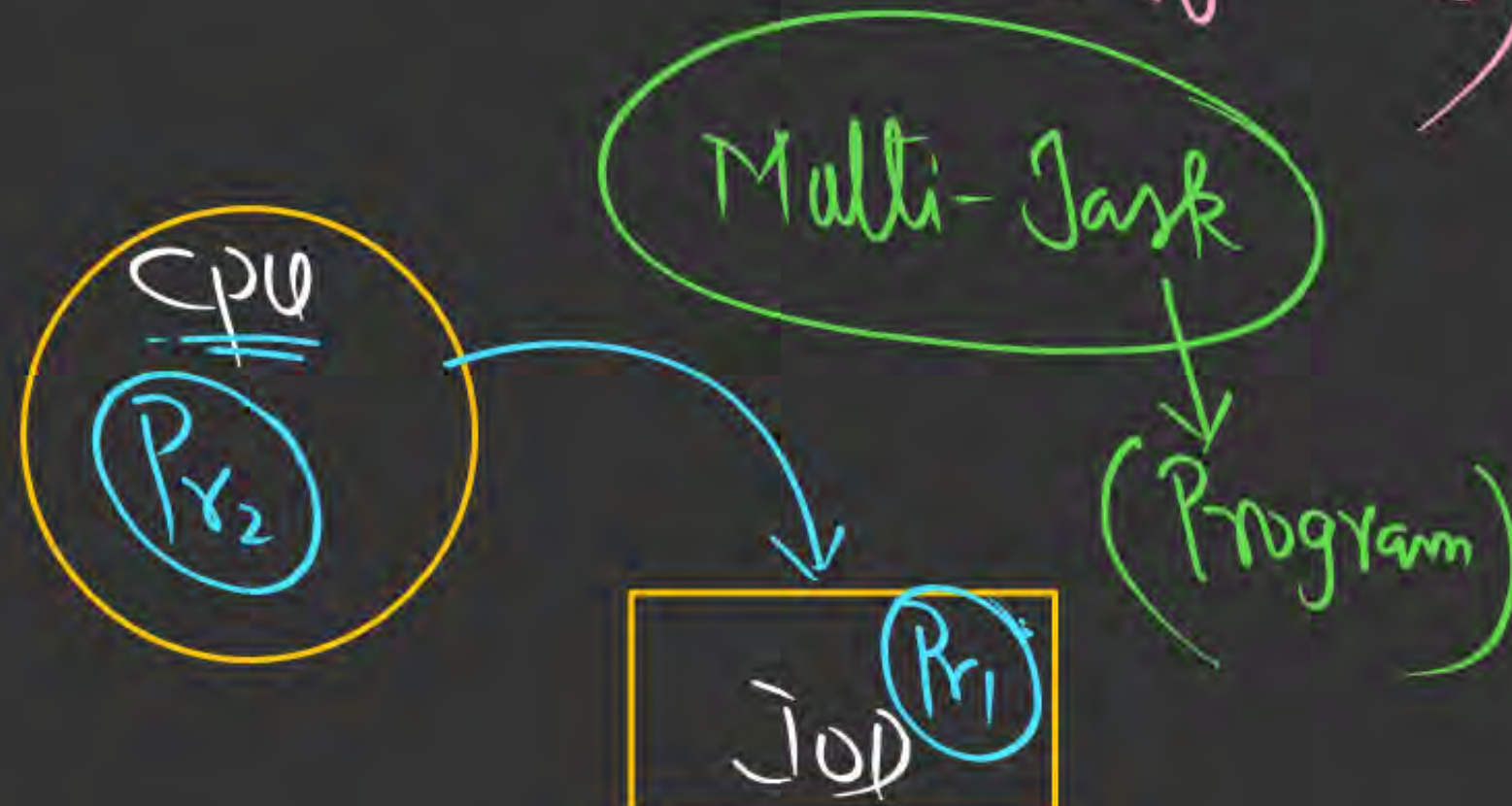
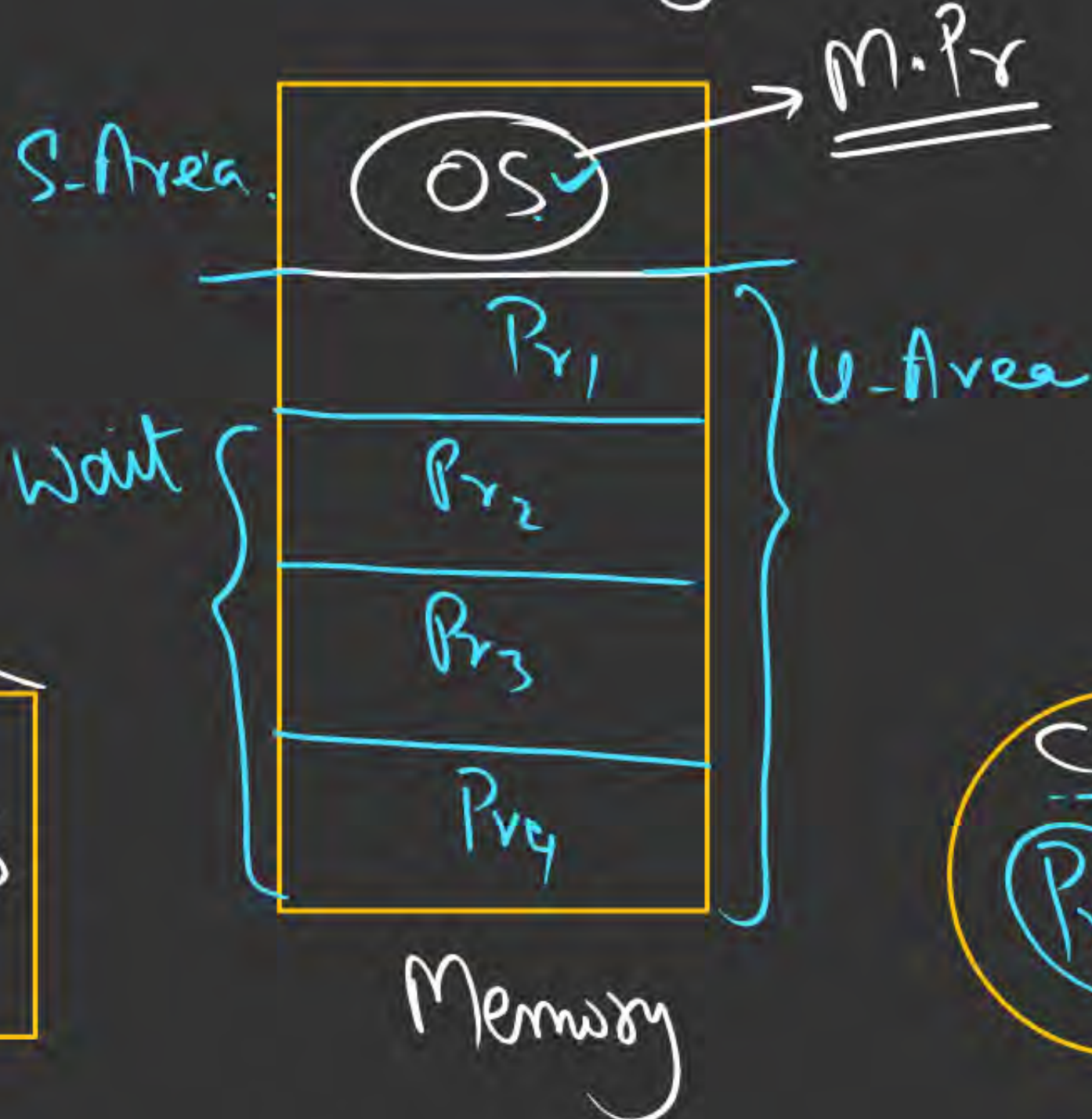
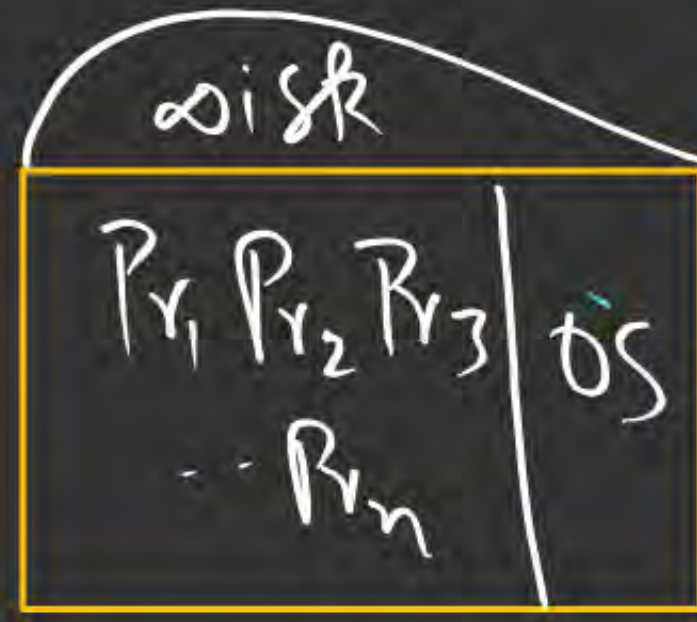
Schematic View of Uniprogrammed O.S

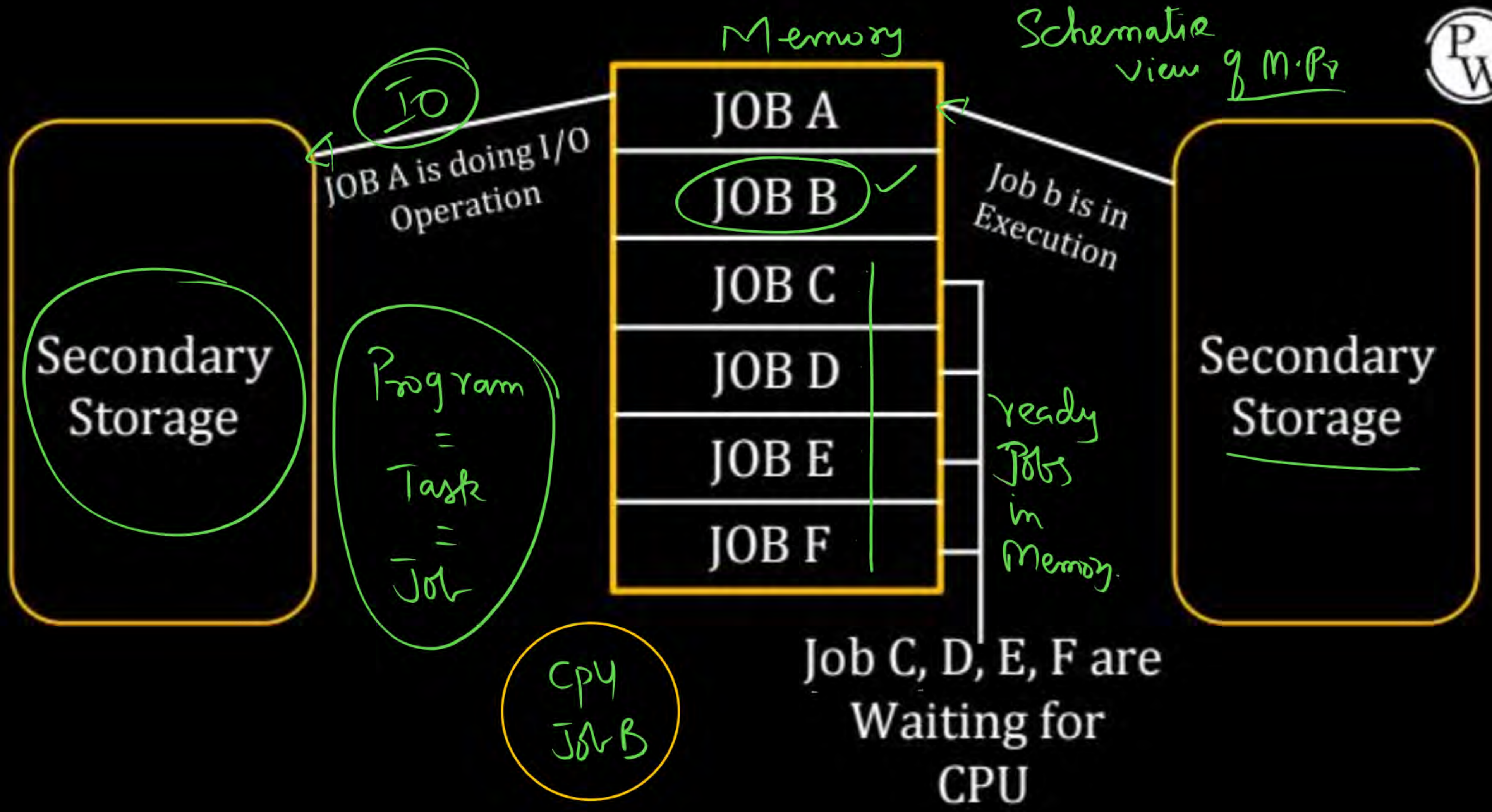


Multi programming: Ability of the O.S to hold (manage) multiple Ready to run Programs in Memory.

↓
(Multiplexing of CPU among different Programs in memory)

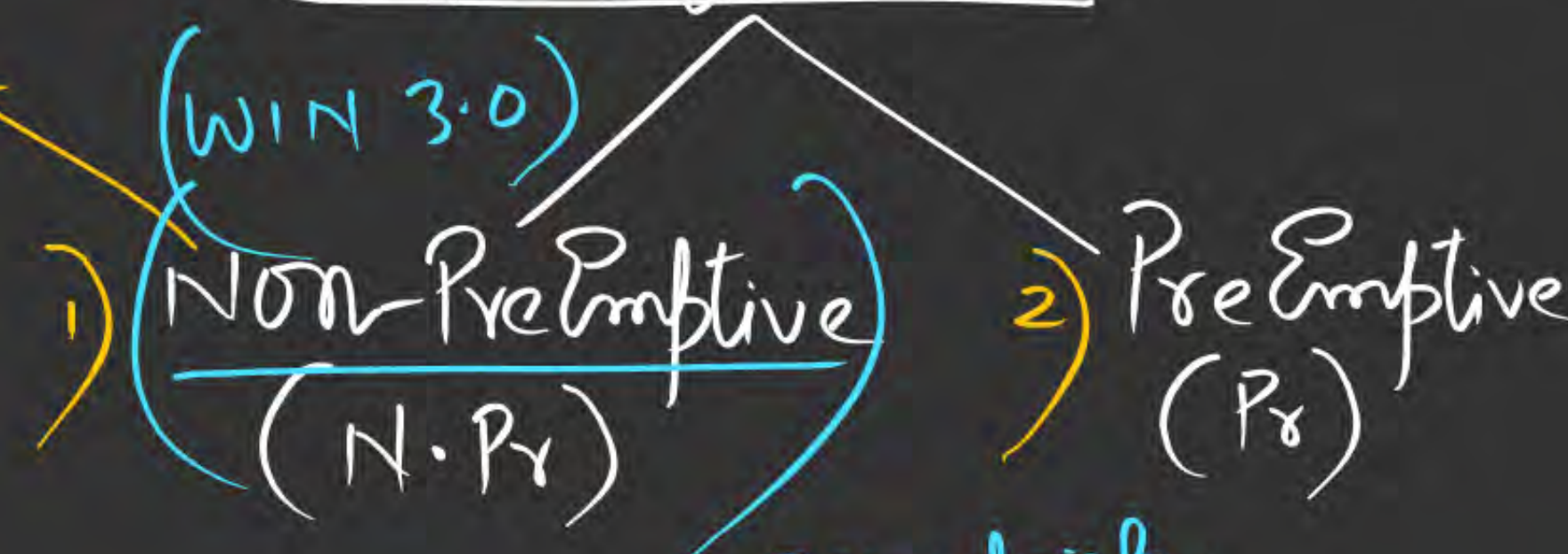
Objective: (Max CPU Utiliz)
Increase Thruput (Efficiency)





* Types of M.Pr.

No forceful de alloc.



PreEmptive : Forceful deallocation

Non-Pr: running Program on cpu will not be forced to leave the cpu. But it will release the cpu voluntarily (on its own)

drawback:
 → Starvation to other waiting Programs

→ Lack of Interactiveness (i) Complete its execution (ii) Needs IO/Sys Call



PreEmptive M.P.r. O.S (Objective) (Improve Interactiveness, responsiveness)

WIN-95

(WIN-10)

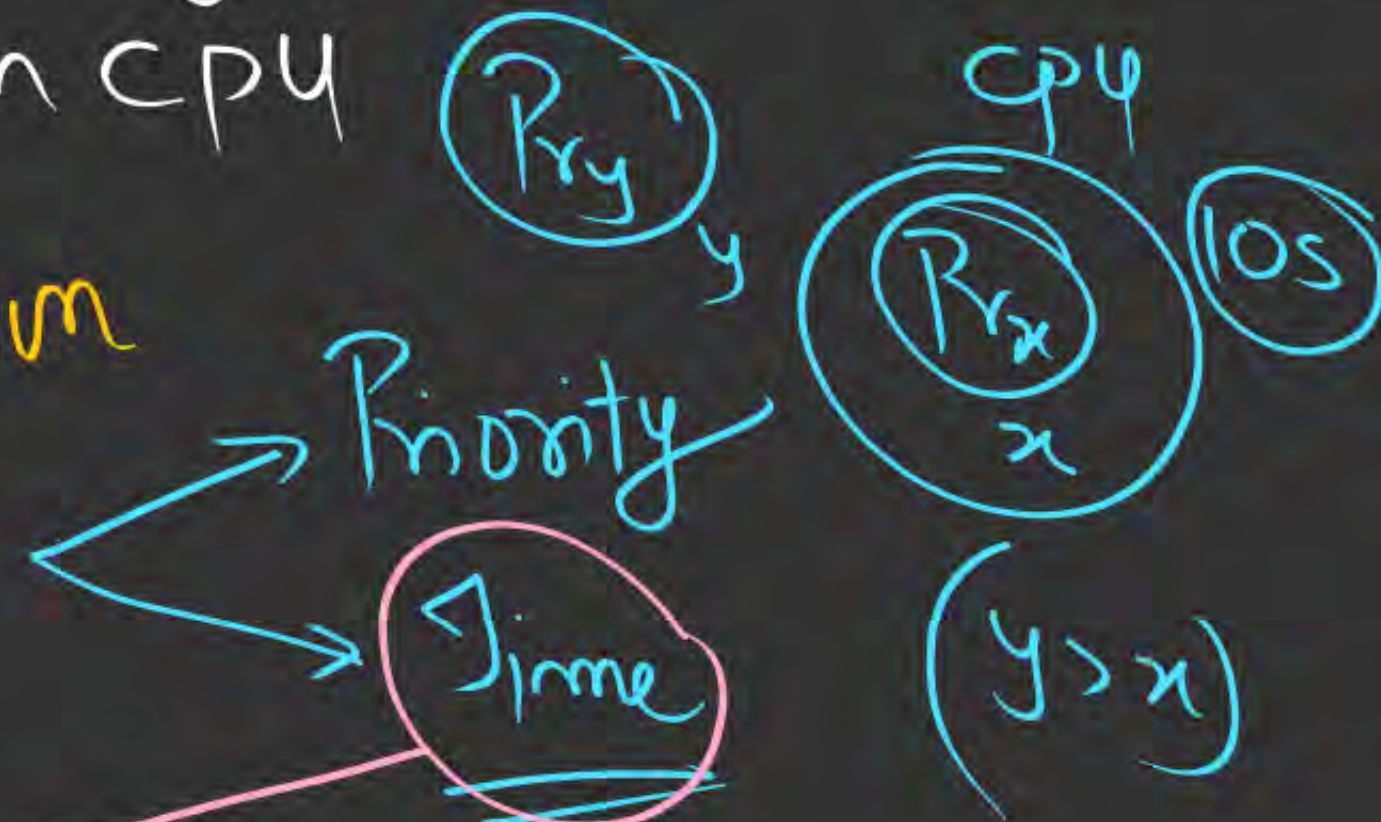
UNIX

LINUX

MAC

: Program running on CPU, can get forcefully deallocated from CPU (PreEmption) ^(OS), So that other waiting Programs can get their chance to run on CPU

(How in PreEmption done)



(M.P.r. Timeshared OS)

<UNIX> Multi programming vs Multi-Tasking <WIN-MS>

Objective

→ Study the design & Implementation / Policies

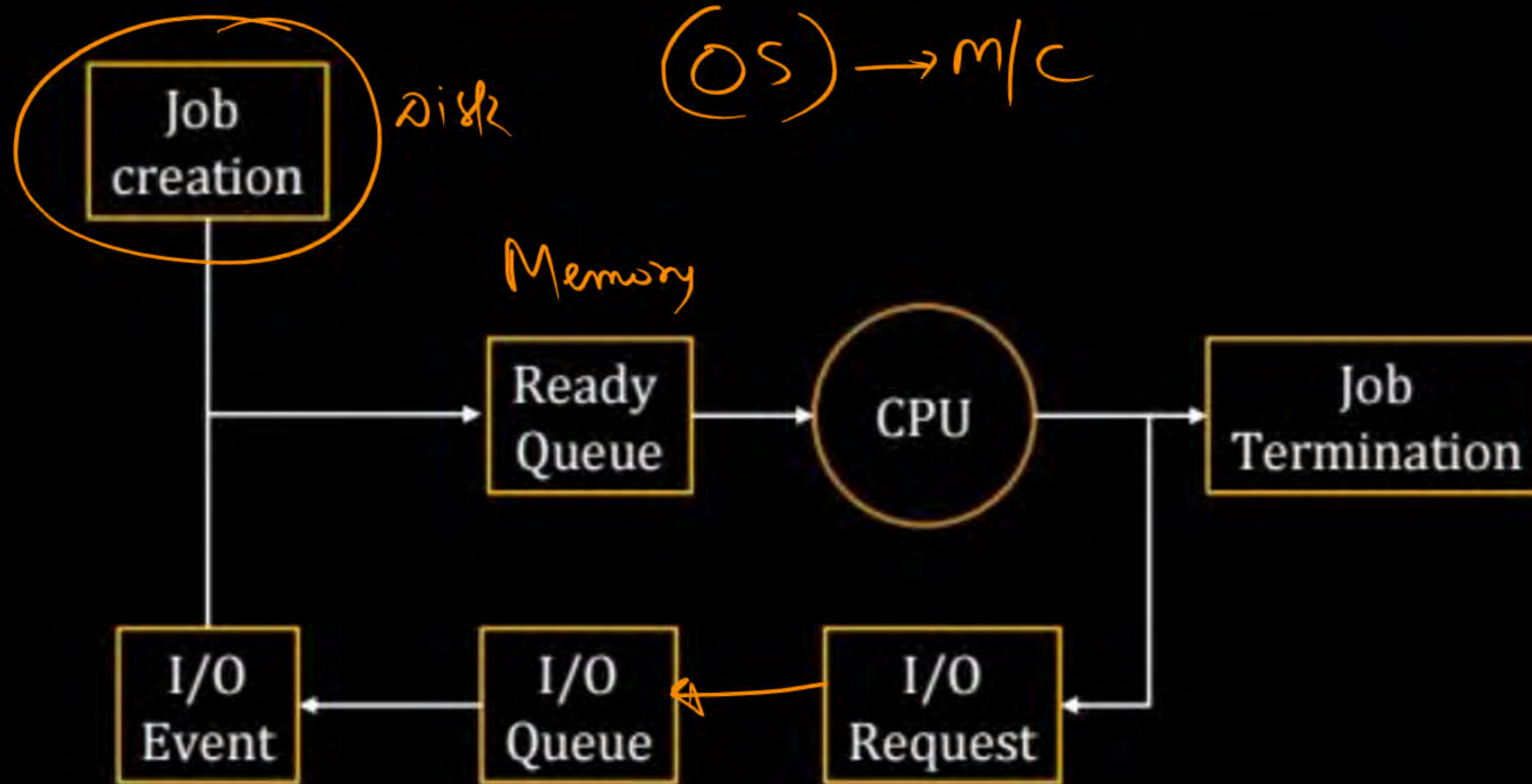
of a Pre-emptive based

M. Pr. O. S (like UNIX/LINUX/WIN/MAC)

Pre-emptive based
M. Pr. O. S

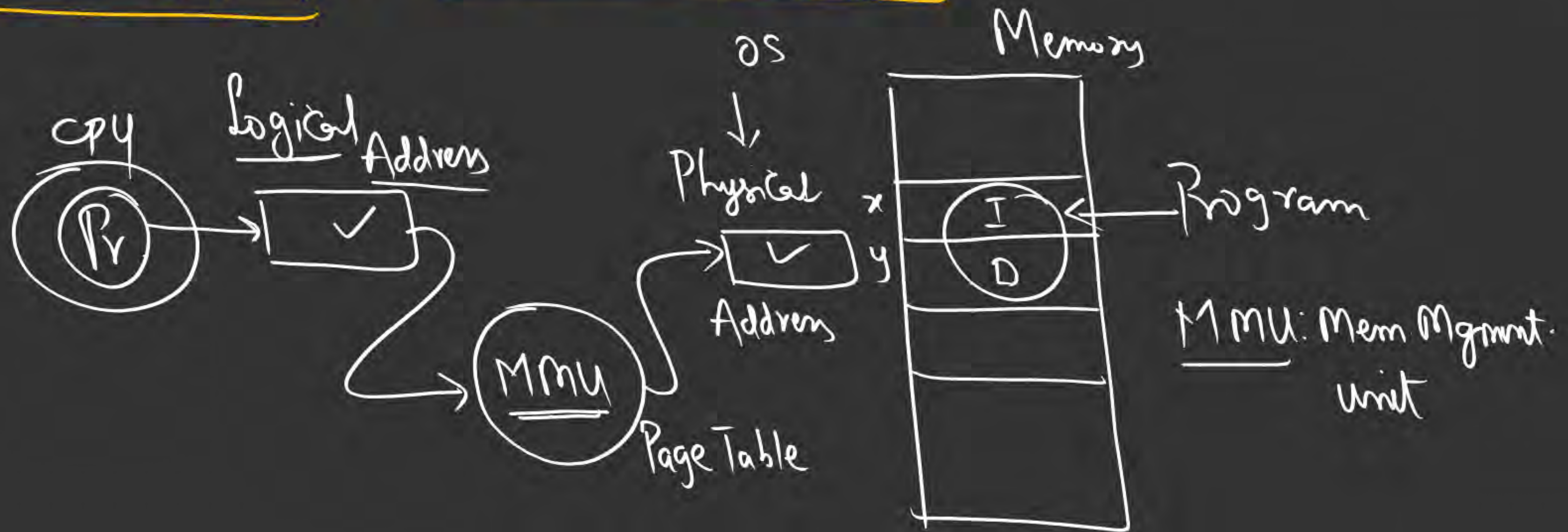
Multiprogramming

A Schematic view of Multiprogramming

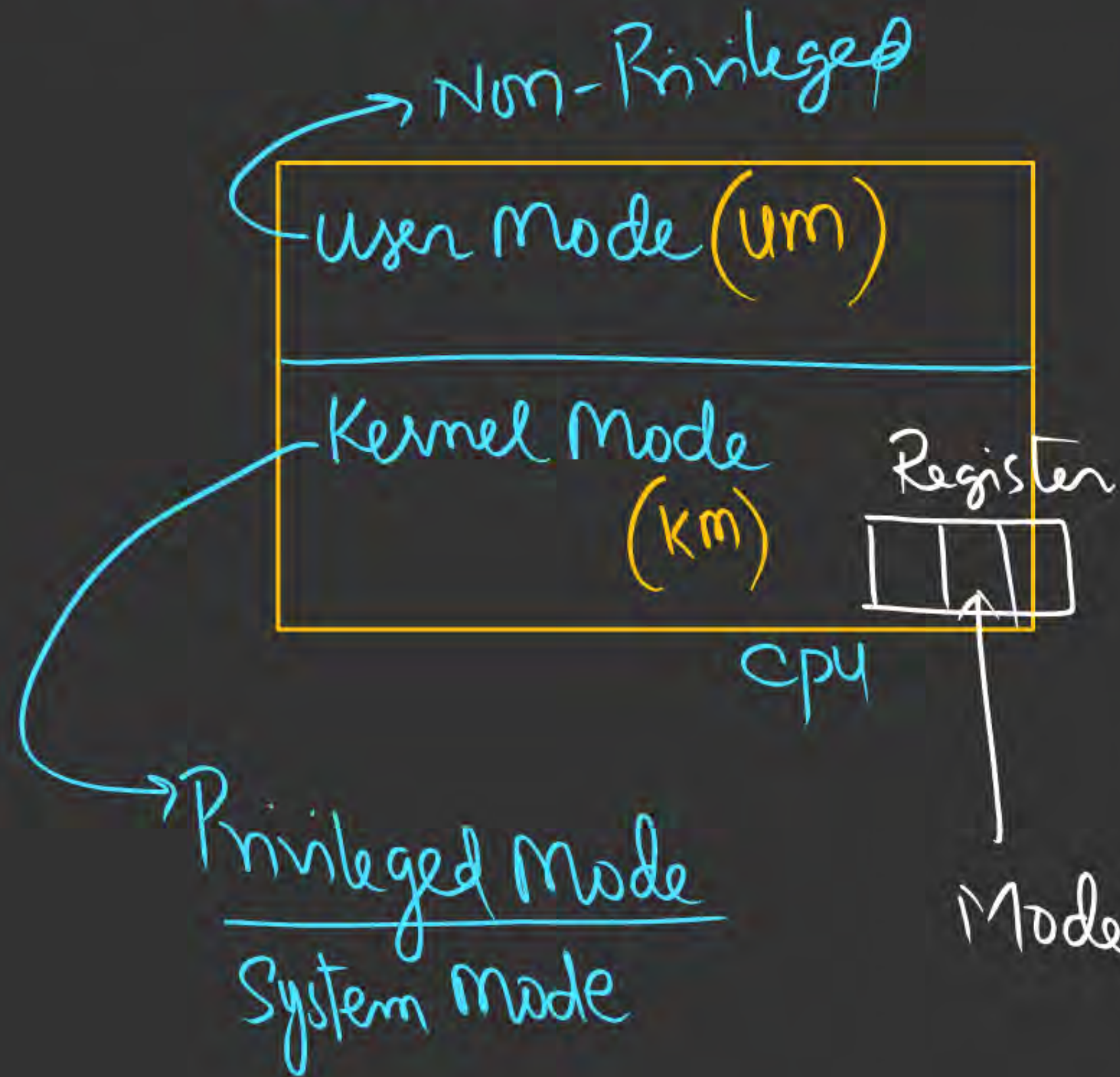


* Architectural (H/w) Requirements for Implementing a M.Pr. O.S

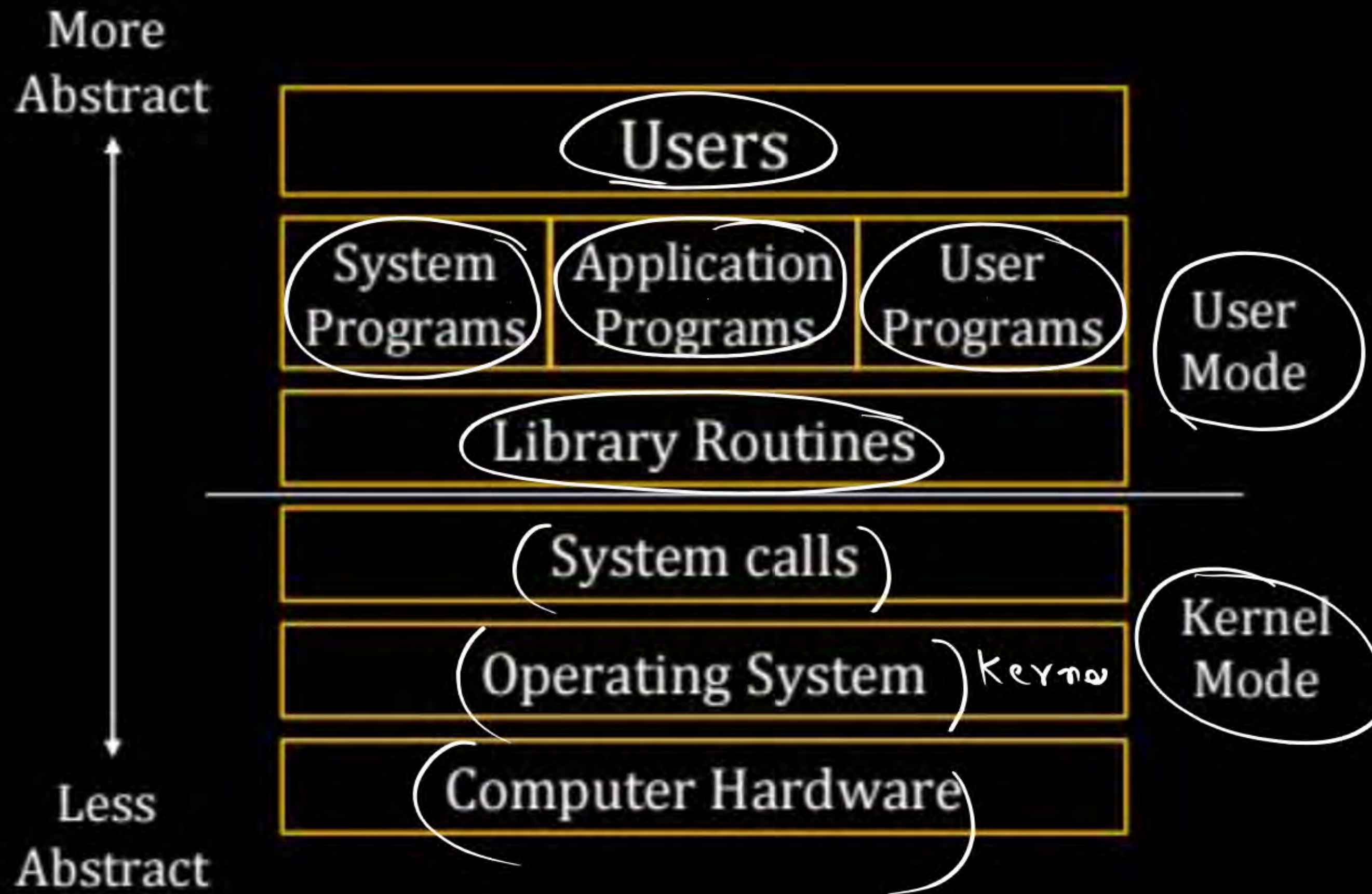
- ① IO (Sec-Storage services) : DMA [Direct Memory Access] < CPU; Memory; IO (Sec-Storage) >
- ② Memory (RAM) : Address Translation



(*) 3) CPU : Dual mode operation :
Processor Two



um	km
1) All user appl. run in um	1) All OS routines Programs run in km
2) um is PreEmptive (Non-Atomic)	2) km is non-PreEmptive (Atomic)
3)	



→ Many a Times, it becomes necessary to shift the Mode from U → K & back from K → U;

Mode Shifting
(is needed to avail OS - services)

(??)

